

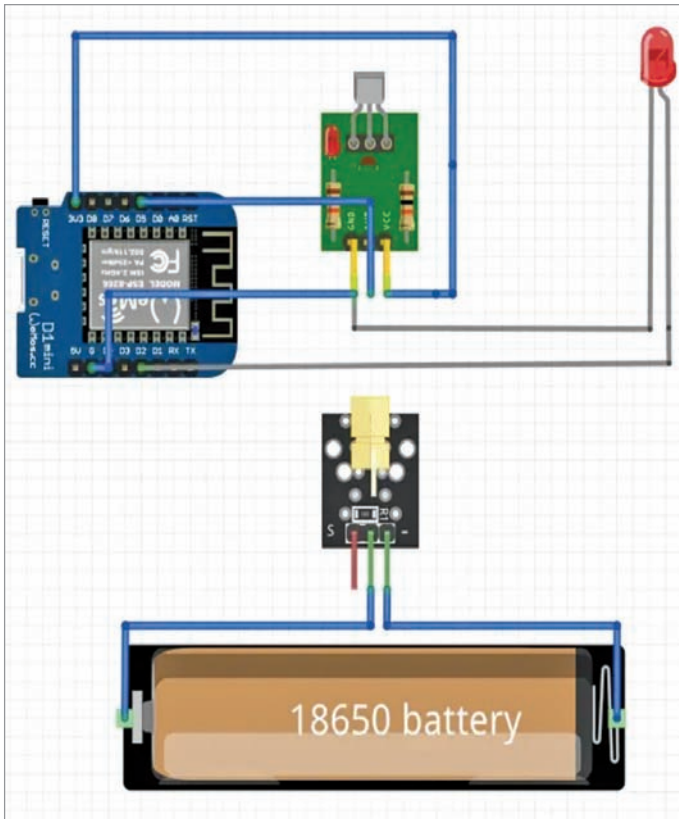


Fig. 8: Wiring diagram of the transmitter and receiver

```
function doGet(e) {
  Logger.log(JSON.stringify(e));
  var result = 'OK';

  if (!e.parameter || !e.parameter.laserState) {
    result = 'No Parameters';
  } else {
    var sheet_id = '1S...';
    var sheet = SpreadsheetApp.openById(sheet_id);
    var newRow = sheet.getLastRow() + 1;
    var rowData = [];
    var Curr_Date = new Date();
    rowData[0] = Curr_Date; // Date in column A
    var Curr_Time = Utilities.formatDate(Curr_Date, 'GMT-05:00', 'MM/dd/yyyy hh:mm:ss');
    rowData[1] = Curr_Time; // Time in column B

    var laserState = stripQuotes(e.parameter.laserState);
    Logger.log('Laser State: ' + laserState);

    // Store the laser state in column C
    rowData[2] = laserState;

    Logger.log(JSON.stringify(rowData));
    var newRange = sheet.getRange(newRow, 1, 1, 3);
    newRange.setValues([rowData]);

    result = 'OK';
  }
}
```

Fig. 9: Google Sheet script

resistor (10kΩ) to ensure the output is collected. The module operates on a 5V DC power supply, and its output can only go low (L). The open-collector transistor can sink a maximum of 20mA of current.

Circuit and working

Fig. 3 shows the circuit diagram of the device. The setup is based on the WeMos D1 Mini (MOD1), light sensor receiver (S1), and laser light transmitter. The receiver unit can be powered with a 5V adaptor after uploading the source code, if the 3.7V battery is not used. The transistor is powered by the 18650 rechargeable battery (3.7V). The internal assembly of the laser light sensor is shown in Fig. 4, while Fig. 5 provides the pin details of the WeMos board.

First, create the Google Sheet

and obtain its ID. This can be found in the URL of the Google Sheet, as shown in Fig. 6. Once the sheet ID is noted, proceed to configure the Wi-Fi SSID and password for the local network. Next, configure the URL and Google Sheet ID within the source code.

The sensor data is checked in a continuous loop and updated on the

Google Sheet. Fig. 7 shows a snippet of the source code.

Testing

Begin by assembling the components, referring to Figs. 4 and 8 before completing the connections. Ensure the code is uploaded to the ESP8266 chip (WeMos D1 Mini board). Connect the RX and TX (laser receiver and transmitter) circuits as shown in the wiring diagram in Fig. 8.

Once the components are correctly assembled, verify the connections. Power the TX laser device and place the RX device in the line of sight. **EFY**

EFY Note
The source code
of this project
is available
for free download at
www.electronicsforu.com

